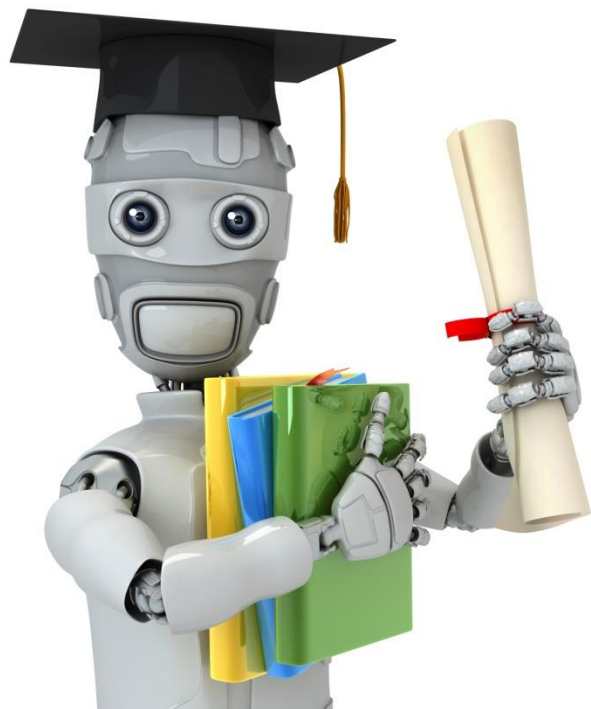




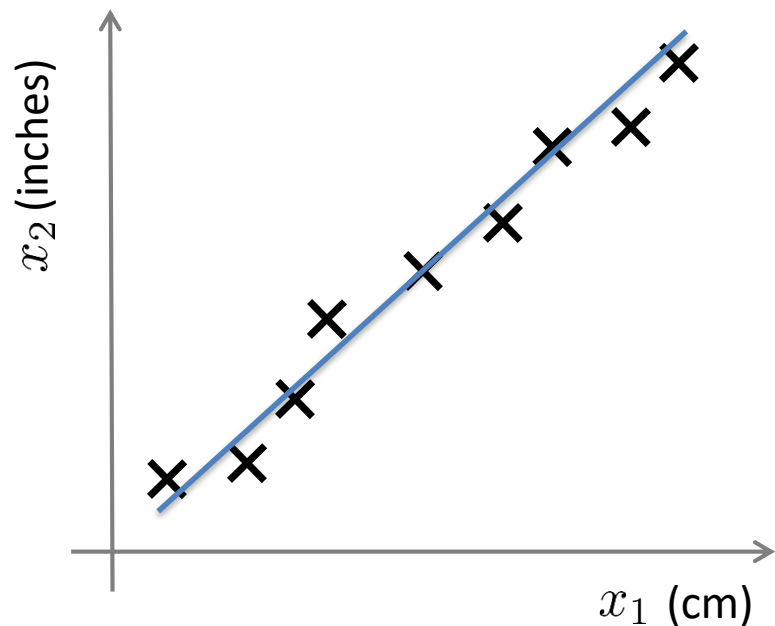
Machine Learning

Dimensionality Reduction

Motivation I: Data Compression

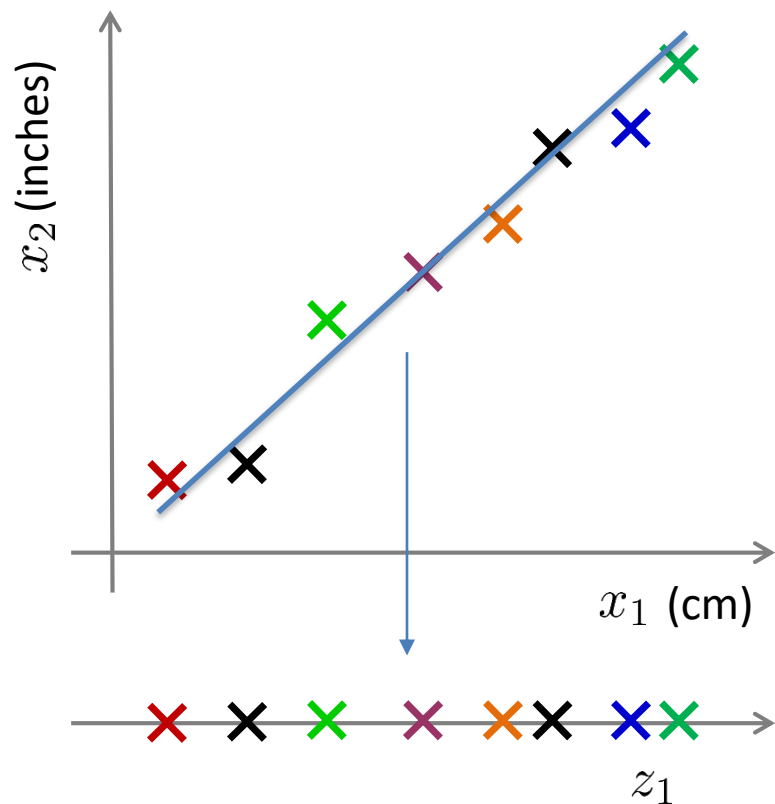
特征选择：直接从较多的特征中选取重要的特征
特征降维：通过特征投影后再筛选出重要的特征
(重要性排序)

Data Compression



Reduce data from
2D to 1D

Data Compression



Reduce data from
2D to 1D

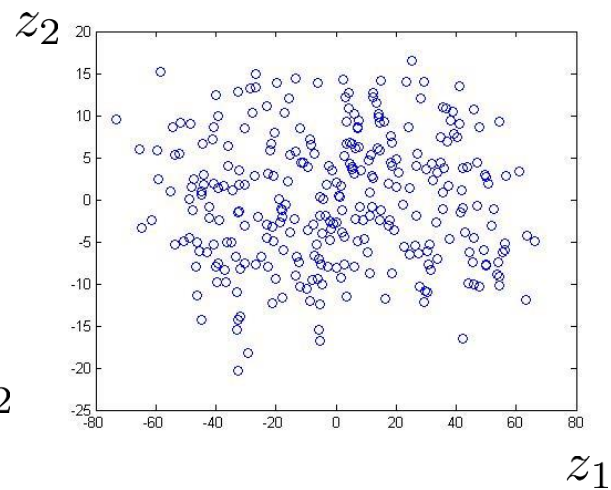
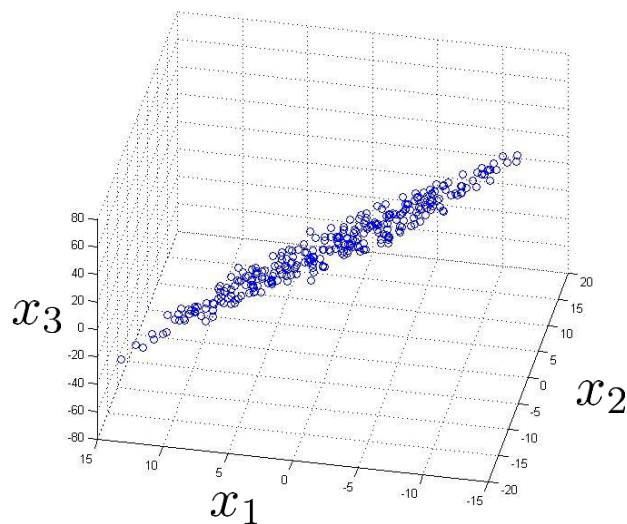
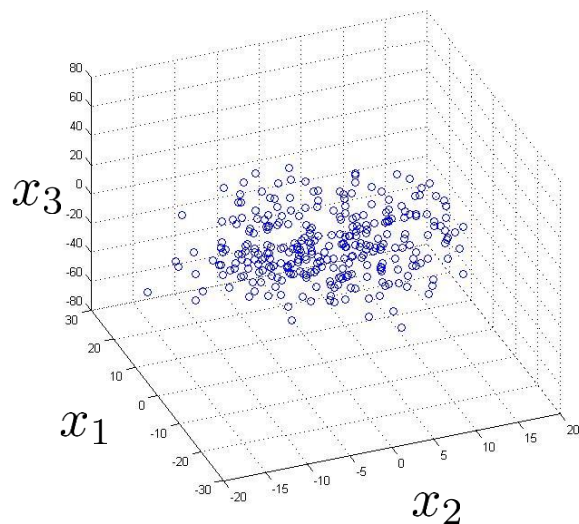
$$x^{(1)} \rightarrow z^{(1)}$$

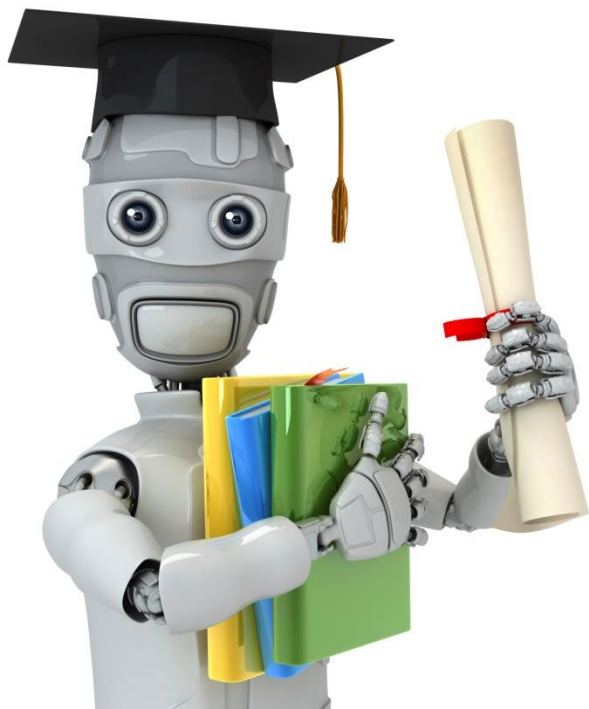
$$x^{(2)} \rightarrow z^{(2)}$$

$$x^{(m)} \rightarrow z^{(m)}$$

Data Compression

Reduce data from 3D to 2D





Machine Learning

Dimensionality Reduction

Motivation II: Data Visualization

Data Visualization

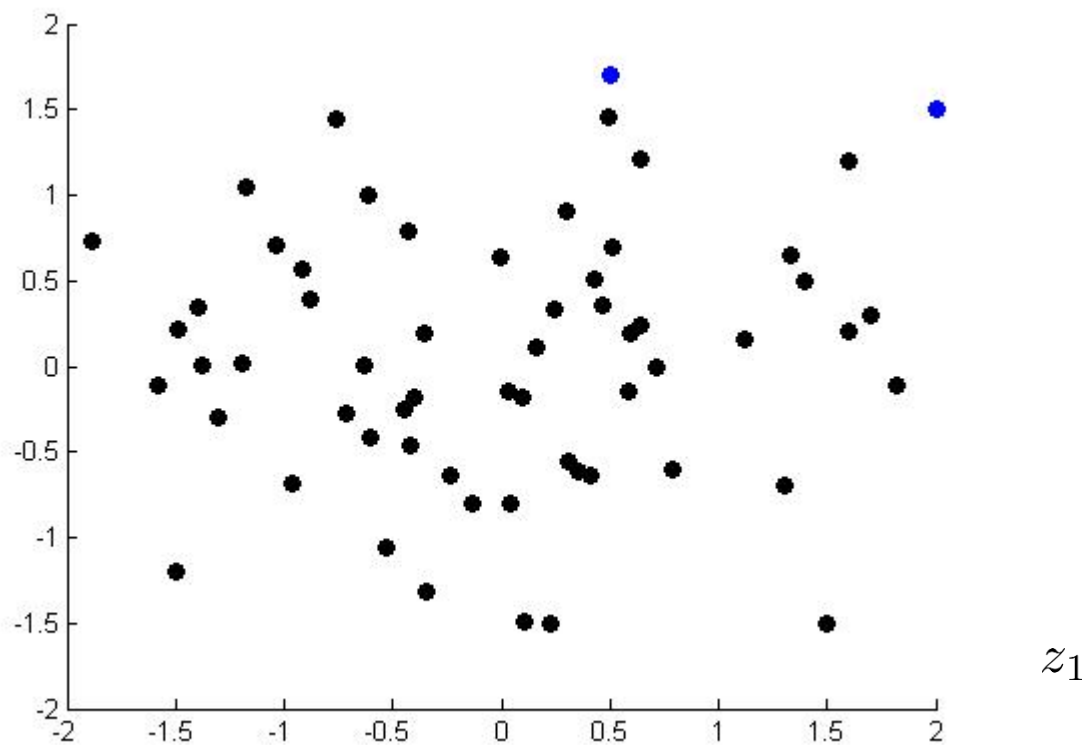
Country	GDP (trillions of US\$)	Per capita GDP (thousands of intl. \$)	Human Develop- ment Index	Life expectancy	Poverty Index (Gini as percentage)	Mean household income (thousands of US\$)	...
Canada	1.577	39.17	0.908	80.7	32.6	67.293	...
China	5.878	7.54	0.687	73	46.9	10.22	...
India	1.632	3.41	0.547	64.7	36.8	0.735	...
Russia	1.48	19.84	0.755	65.5	39.9	0.72	...
Singapore	0.223	56.69	0.866	80	42.5	67.1	...
USA	14.527	46.86	0.91	78.3	40.8	84.3	...
...	...	...	...	...	...	...	...

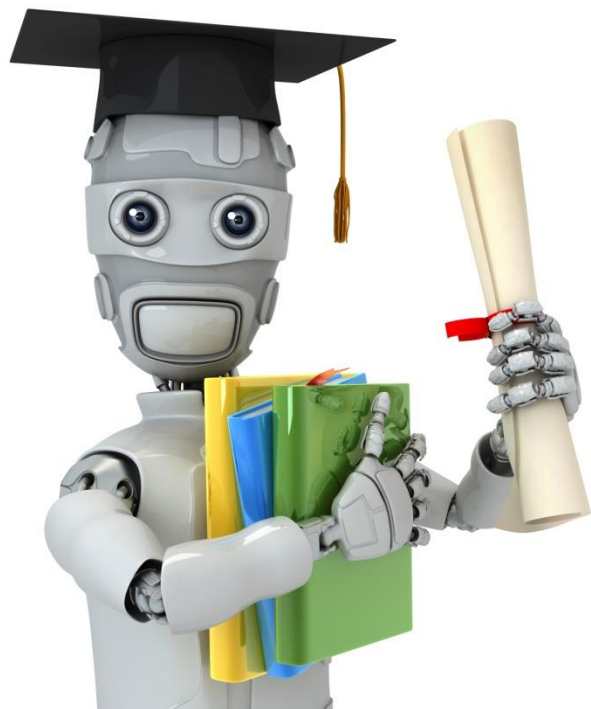
Data Visualization

将多维数据降到3维以内进行可视化

Country	z_1	z_2
Canada	1.6	1.2
China	1.7	0.3
India	1.6	0.2
Russia	1.4	0.5
Singapore	0.5	1.7
USA	2	1.5
...	...	...

Data Visualization





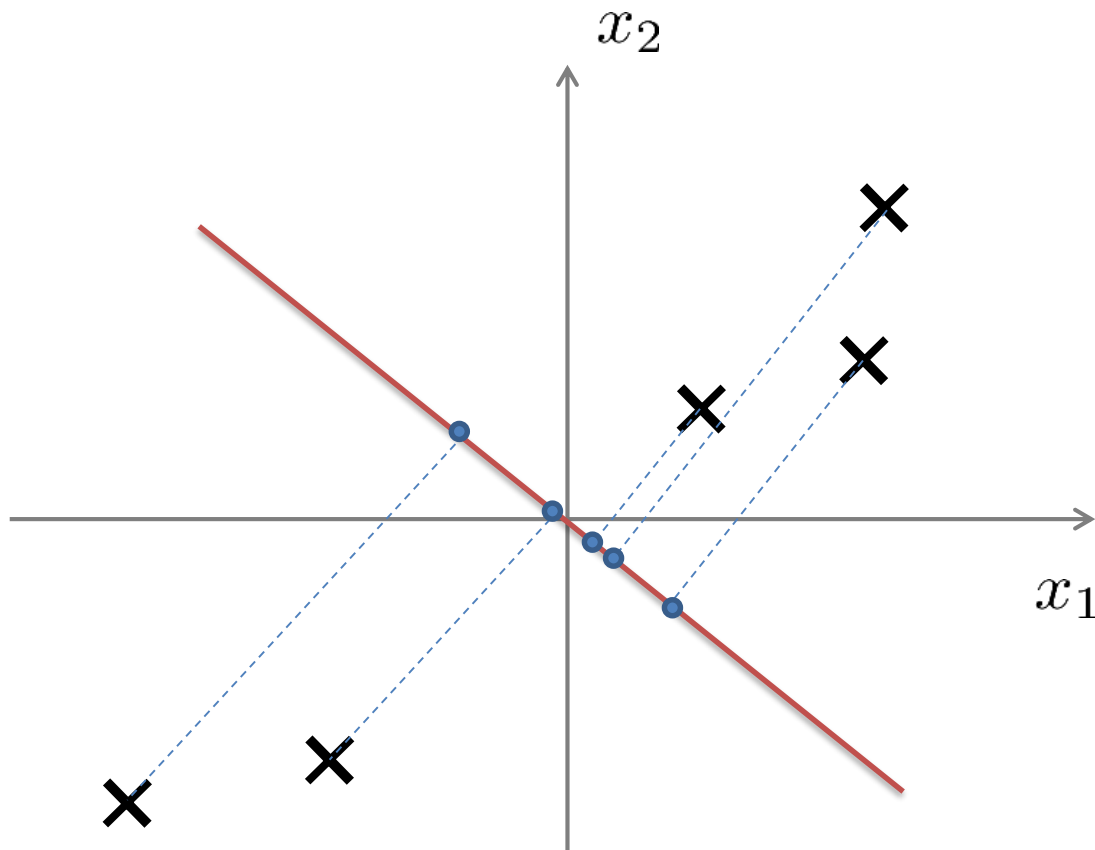
Machine Learning

Dimensionality Reduction

Principal Component
Analysis problem
formulation

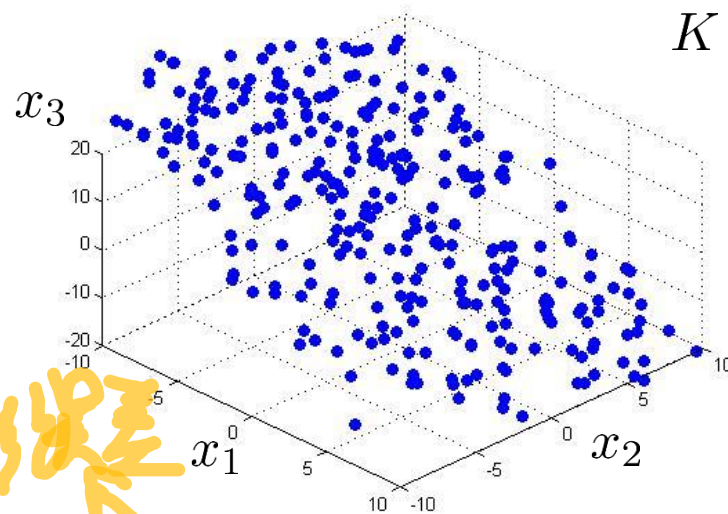
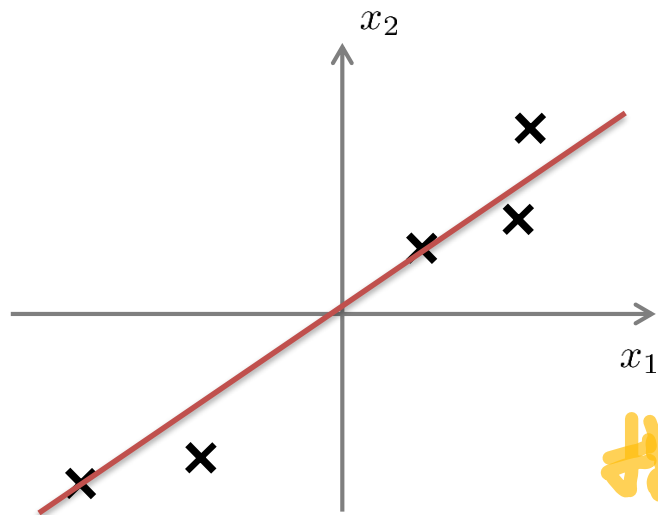
PCA

Principal Component Analysis (PCA) problem formulation



Principal Component Analysis (PCA) problem formulation

$$\begin{aligned} 3D &\rightarrow 2D \\ K &= 2 \end{aligned}$$

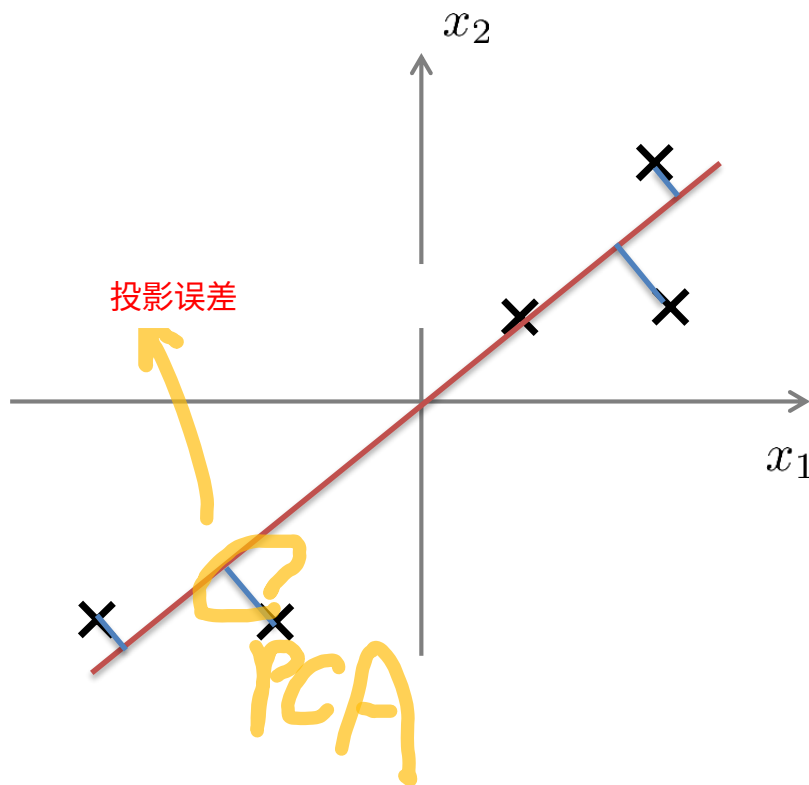
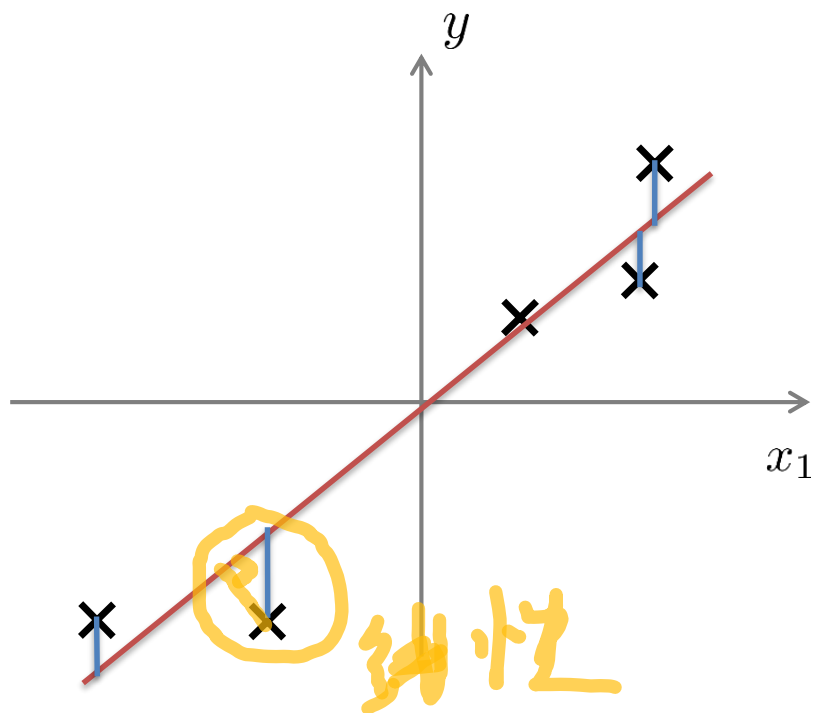


投影误差

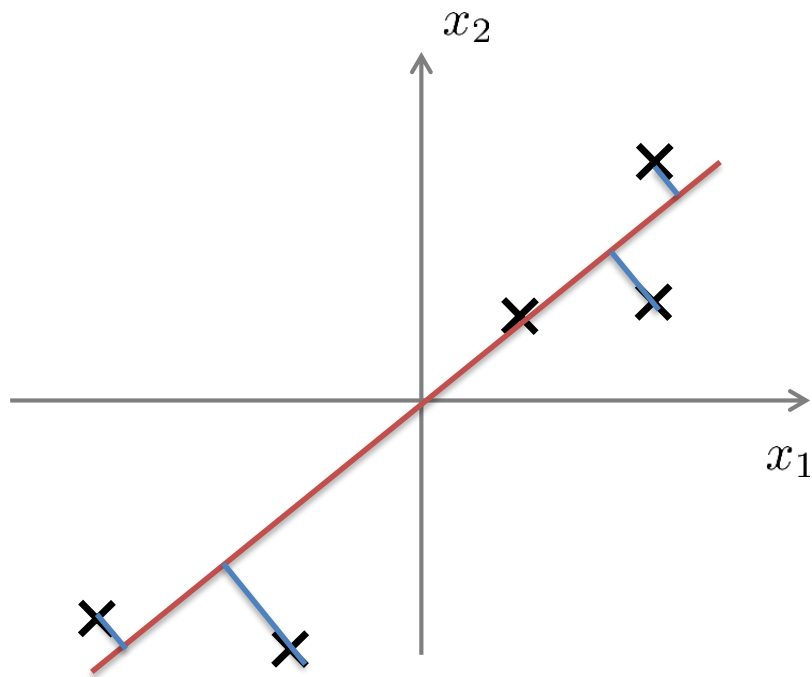
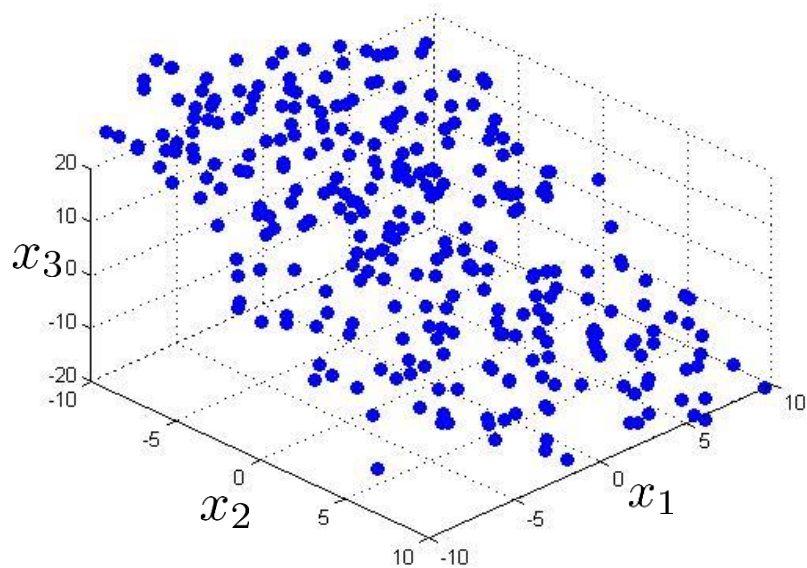
Reduce from 2-dimension to 1-dimension: Find a direction (a vector $u^{(1)} \in \mathbb{R}^n$) onto which to project the data so as to minimize the projection error 方差保留多，区分度高

Reduce from n -dimension to k -dimension: Find k vectors $u^{(1)}, u^{(2)}, \dots, u^{(k)}$ onto which to project the data, so as to minimize the projection error.

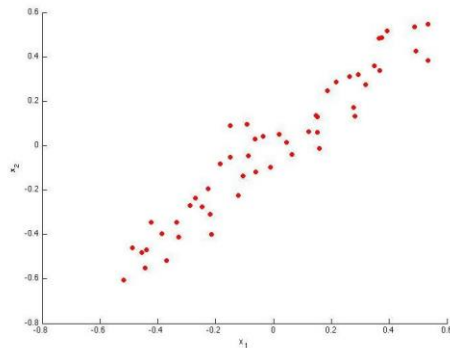
PCA is not linear regression



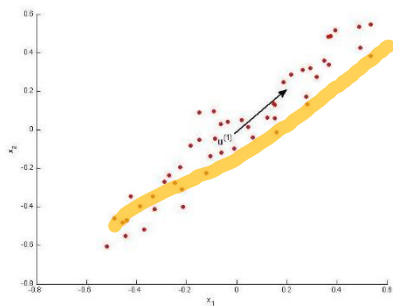
PCA is not linear regression



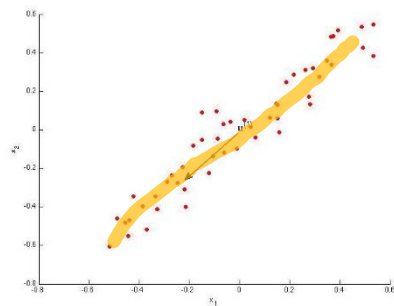
考虑以下二维数据集：



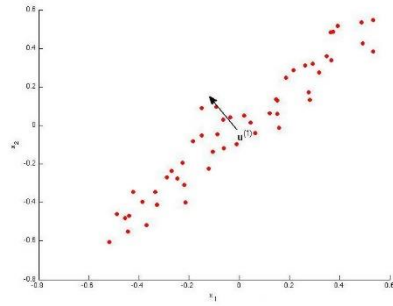
下列哪个图片对应的PCA可能返回的 $u^{(1)}$ （第一特征向量/第一主成分）的值？选出所有正确项



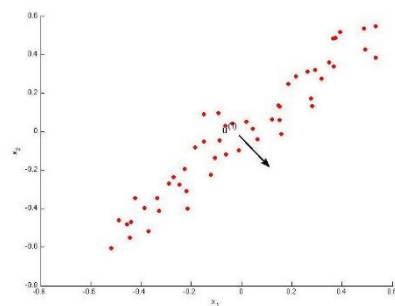
A



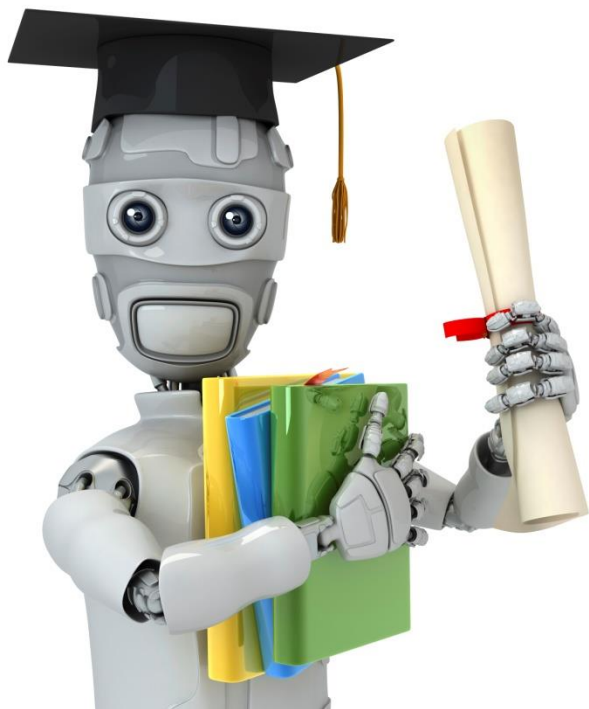
B



C



D



Machine Learning

Dimensionality Reduction

Principal Component Analysis algorithm

Data preprocessing

Training set: $x^{(1)}, x^{(2)}, \dots, x^{(m)}$

Preprocessing (feature scaling/mean normalization):

$$\mu_j = \frac{1}{m} \sum_{i=1}^m x_j^{(i)}$$

⇒ 计算均值点

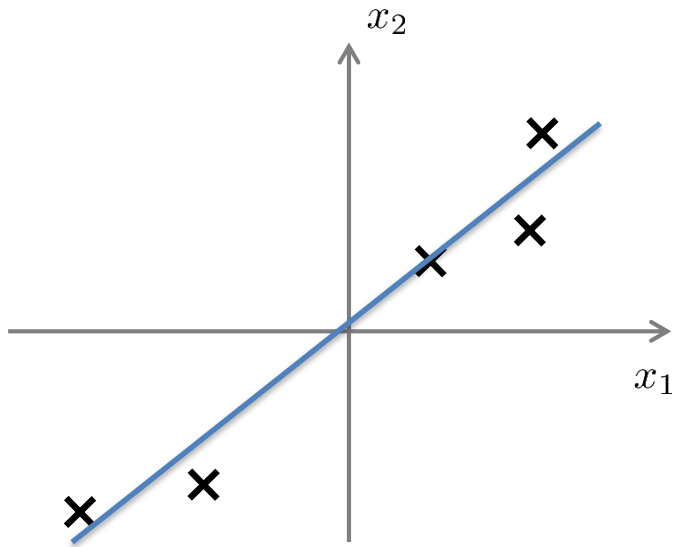
Replace each $x_j^{(i)}$ with $x_j - \mu_j$

If different features on different scales (e.g., x_1 = size of house, x_2 = number of bedrooms), scale features to have comparable range of values.

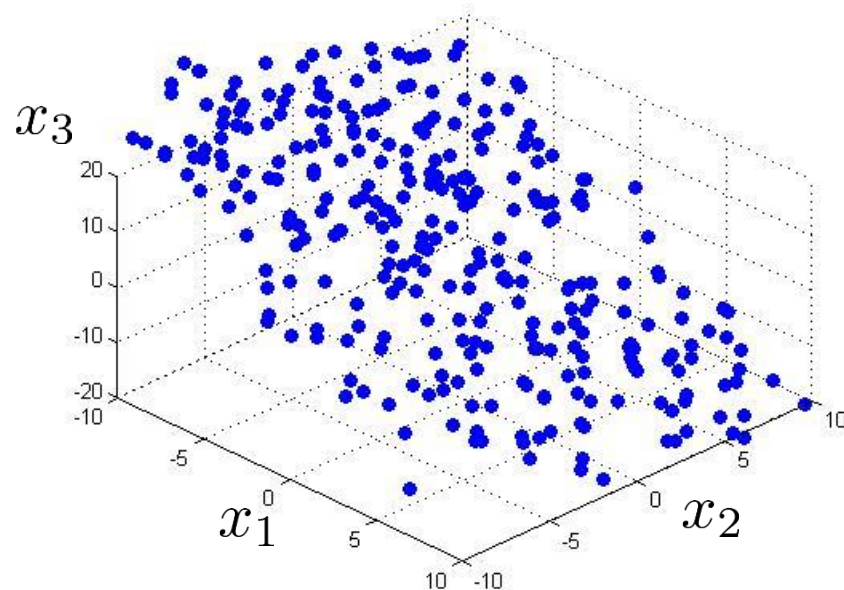
$$x_j = \frac{x_j - \mu_j}{s_j}$$

⇒ 归一化

Principal Component Analysis (PCA) algorithm



Reduce data from 2D to 1D



Reduce data from 3D to 2D

Principal Component Analysis (PCA) algorithm

Reduce data from n -dimensions to k -dimensions

Compute "covariance matrix":

$$\Sigma = \frac{1}{m} \sum_{i=1}^n (x^{(i)})(x^{(i)})^T$$

协方差

$n \times n$ 矩阵，反映不同维度之间的相关性

Compute "eigenvectors" of matrix Σ :

$$[U, S, V] = \text{svd}(\text{Sigma});$$

特征向量

$n \times n$ Matrix

$$U = \begin{bmatrix} | & | & & | \\ u^{(1)} & u^{(2)} & \dots & u^{(n)} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

奇异值分解求特征向量

$n \times n$ 正交的特征向量矩阵

特征向量

按照特征值大小递减排序，筛选重要的特征向量（特征），从而保留更多的数据信息

Principal Component Analysis (PCA) algorithm

From $[U, S, V] = \text{svd}(\text{Sigma})$, we get:

$$U = \begin{bmatrix} | & | & & | \\ u^{(1)} & u^{(2)} & \dots & u^{(n)} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times n}$$

$$z^{(i)} = U_{reduce}^T * x^{(i)} =$$

$$\underbrace{\begin{bmatrix} | & | & & | \\ u^{(1)} & u^{(2)} & \dots & u^{(k)} \\ | & | & & | \end{bmatrix}}_{n \times k \text{ Matrix}}^T * x^{(i)} = \underbrace{k \times 1 \text{ Matrix}}_{\text{降维后的k维特征}}$$

选k个特征向量

$n \times k$ Matrix

$n \times 1$ Matrix

Principal Component Analysis (PCA) algorithm summary

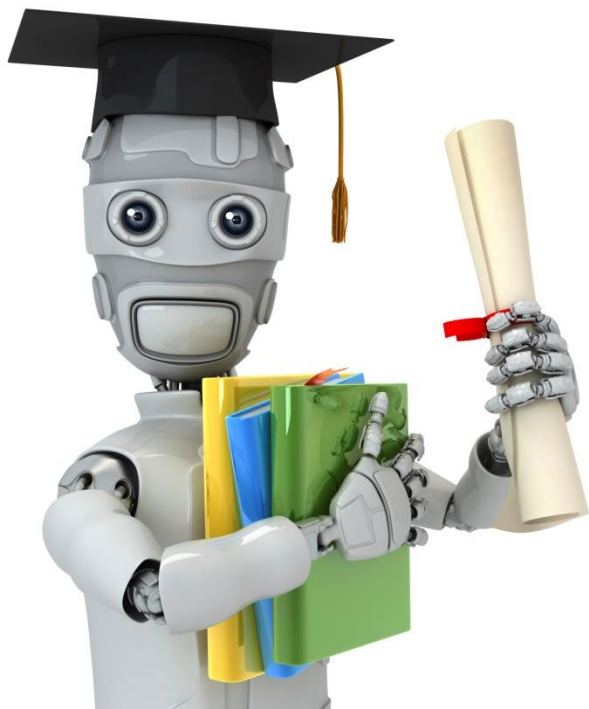
After mean normalization (ensure every feature has zero mean) and optionally feature scaling:

$$\mathbf{Sigma} = \frac{1}{m} \sum_{i=1}^m (x^{(i)})(x^{(i)})^T$$

```
[U,S,V] = svd(Sigma);
```

```
Ureduce = U(:,1:k);
```

```
z = Ureduce'*x;
```

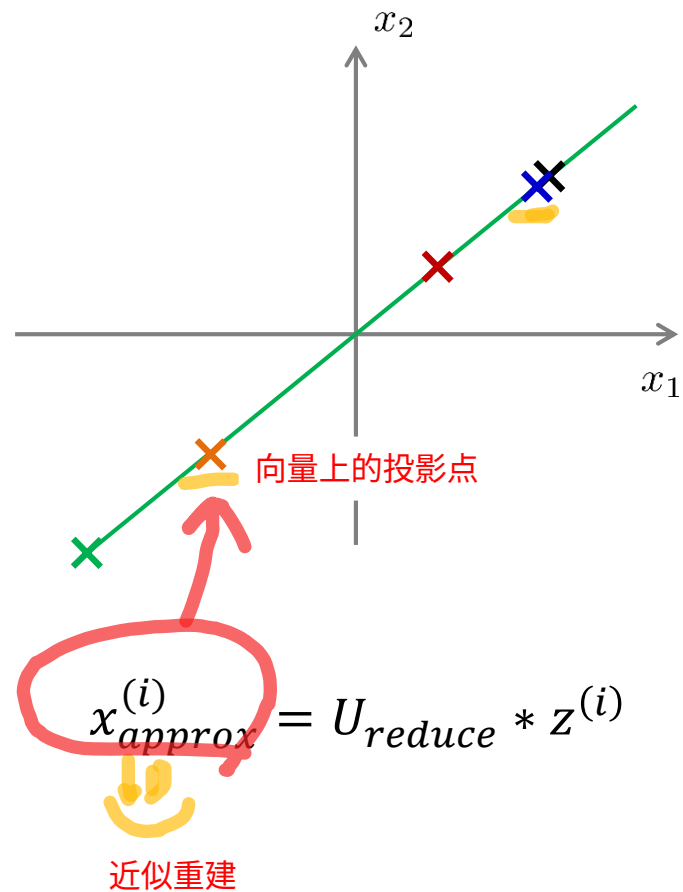
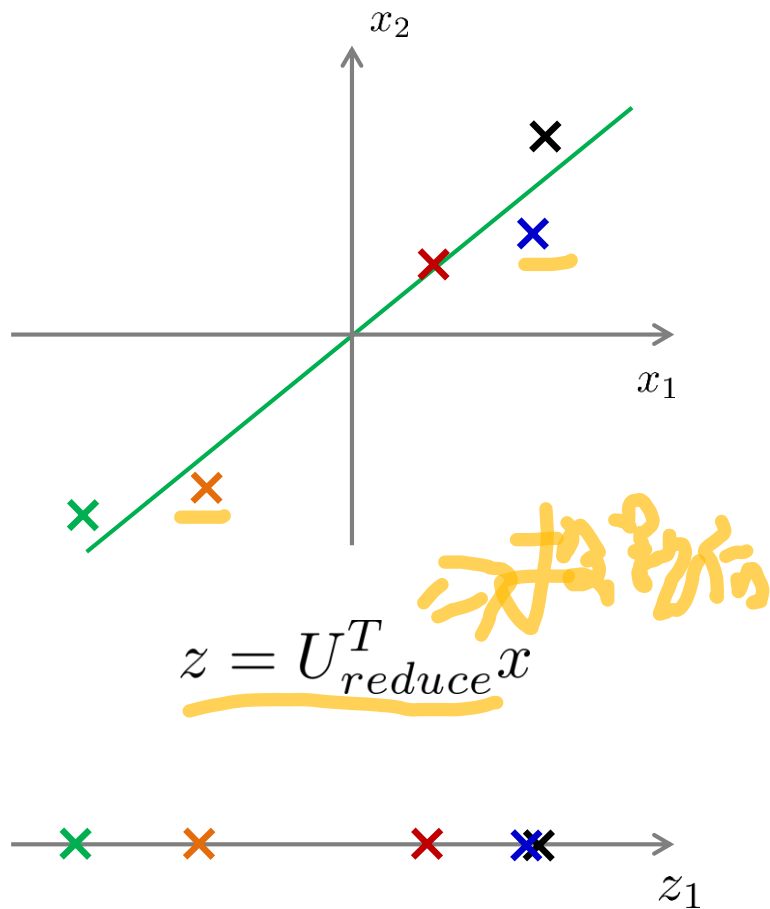


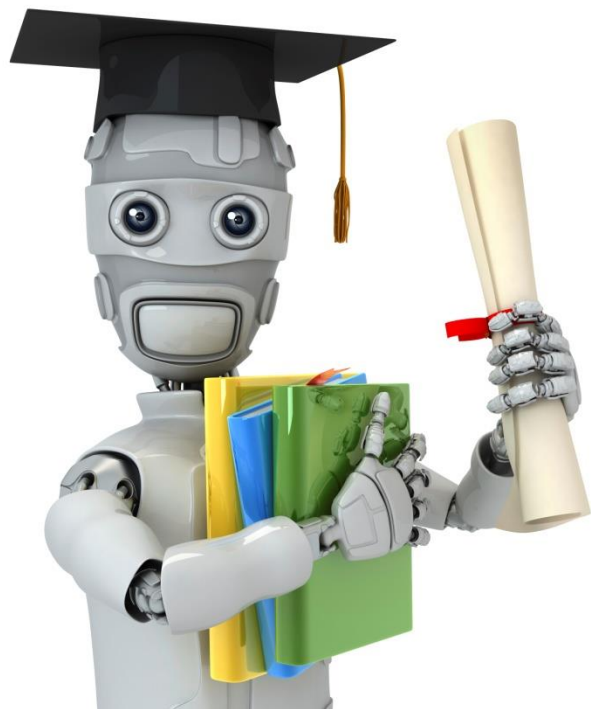
Machine Learning

Dimensionality Reduction

Reconstruction from
compressed
representation

Reconstruction from compressed representation





Machine Learning

Dimensionality Reduction

Choosing the number of principal components

选择主成分的数量

Choosing k (number of principal components)

Average squared projection error: $\frac{1}{m} \sum_{i=1}^m \|x^{(i)} - x_{approx}^{(i)}\|^2$

Total variation in the data: $\frac{1}{m} \sum_{i=1}^m \|x^{(i)}\|^2$ 平均投影误差

Typically, choose k to be smallest value so that

$$\frac{\frac{1}{m} \sum_{i=1}^m \|x^{(i)} - x_{approx}^{(i)}\|^2}{\frac{1}{m} \sum_{i=1}^m \|x^{(i)}\|^2} \leq 0.01 \quad (1\%)$$

数据集的方差

选取最小的k使得保留99%的方差

“99% of variance is retained”

保留的方差

Choosing k (number of principal components)

Algorithm:

Try PCA with $k = 1$

Compute $U_{reduce}, z^{(1)}, z^{(2)}, \dots, z^{(m)}, x_{approx}^{(1)}, \dots, x_{approx}^{(m)}$

Check if

$$\frac{\frac{1}{m} \sum_{i=1}^m \|x^{(i)} - x_{approx}^{(i)}\|^2}{\frac{1}{m} \sum_{i=1}^m \|x^{(i)}\|^2} \leq 0.01?$$



$$[U, S, V] = \text{svd}(\text{Sigma})$$

$$S = \begin{bmatrix} S_{11} & 0 & 0 & 0 \\ 0 & S_{22} & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & S_{nn} \end{bmatrix}$$

$$1 - \frac{\sum_{i=1}^k S_{ii}}{\sum_{i=1}^n S_{ii}} \leq 1\%$$

$$\frac{\sum_{i=1}^k S_{ii}}{\sum_{i=1}^n S_{ii}} \geq 0.99$$

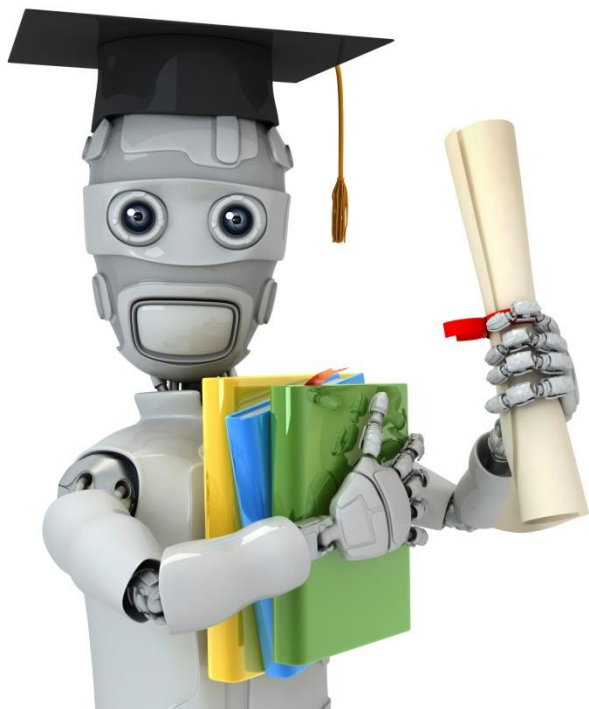
Choosing k (number of principal components)

`[U,S,V] = svd(Sigma)`

Pick smallest value of k for which

$$\frac{\sum_{i=1}^k S_{ii}}{\sum_{i=1}^m S_{ii}} \geq 0.99$$

(99% of variance retained)



Machine Learning

Dimensionality Reduction

Advice for applying PCA

Supervised learning speedup

$$(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})$$

Extract inputs:

Unlabeled dataset: $x^{(1)}, x^{(2)}, \dots, x^{(m)} \in \mathbb{R}^{10000}$

$\downarrow PCA$

$$z^{(1)}, z^{(2)}, \dots, z^{(m)} \in \mathbb{R}^{1000}$$

也需要投影到特征空间

New training set:

$$(z^{(1)}, y^{(1)}), (z^{(2)}, y^{(2)}), \dots, (z^{(m)}, y^{(m)})$$

交叉验证

Note: Mapping $x^{(i)} \rightarrow z^{(i)}$ should be defined by running PCA only on the training set. This mapping can be applied as well to the examples $x_{cv}^{(i)}$ and $x_{test}^{(i)}$ in the cross validation and test sets.

Application of PCA

- Compression
 - Reduce memory/disk needed to store data
 - Speed up learning algorithm
- Visualization

Bad use of PCA: To prevent overfitting

Use $z^{(i)}$ instead of $x^{(i)}$ to reduce the number of features to $k < n$.

Thus, fewer features, less likely to overfit.

解决过拟合问题：PCA不如正则化

⚠ This might work OK, but isn't a good way to address overfitting. Use regularization instead.

$$\min_{\theta} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2m} \sum_{j=1}^n \theta_j^2$$

PCA is sometimes used where it shouldn't be

Design of ML system:

- Get training set $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$
- Run PCA to reduce $x^{(i)}$ in dimension to get $z^{(i)}$
- Train logistic regression on $\{(z^{(1)}, y^{(1)}), \dots, (z^{(m)}, y^{(m)})\}$
- Test on test set: Map $x_{test}^{(i)}$ to $z_{test}^{(i)}$. Run $h_{\theta}(z)$ on $\{(z_{test}^{(1)}, y_{test}^{(1)}), \dots, (z_{test}^{(m)}, y_{test}^{(m)})\}$

How about doing the whole thing without using PCA?

Before implementing PCA, first try running whatever you want to do with the original/raw data $x^{(i)}$. Only if that doesn't do what you want, then implement PCA and consider using $z^{(i)}$.